

Samvardhini Sridharan

Postdoctoral Researcher

University of Washington School of Medicine, Genome Sciences

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Education

Ph.D. in Molecular and Cell Biology with a Designated Emphasis in Genomics and Computational Biology,

University of California – Berkeley

Aug 2020 – Dec 2025

Thesis: *Recurrent Structural Variation and Inversion Haplotype Diversity in Humans and Great Apes*

Thesis supervisor: Dr. Peter Sudmant

Committee members: Drs. Craig Miller, Rasmus Nielsen, Brenna Henn, Priya Moorjani

B.S. in Genetics and Genomics, University of California – Davis

Sep 2016 – Jun 2020

Minors: Statistics and Professional Writing

Research Interests

- Structural variation and chromosomal inversions
- Genome-wide association studies (GWAS) and disease risk
- Primate and comparative genomics
- Population genomics

Research and Industry Experience

Postdoctoral Researcher, University of Washington, Seattle

Mar 2026 – Present

Advisor: Dr. Saori Sakaue

- Developing a genome-wide inversion reference panel and imputation pipeline to enable structural variant association testing across biobank-scale cohorts, including the UK Biobank and *All of Us*.
- Constructing haplotype-resolved assembly alignment pipelines (minimap2, Snakemake) against the CHM13 reference to genotype inversion loci across HGSVC and HPRC cohorts.
- Collaborating with the Eichler lab on inversion callset validation and orthogonal confirmation of ambiguous haplotype calls.

Graduate Student Researcher, University of California, Berkeley

Aug 2020 – Dec 2025

Advisor: Dr. Peter Sudmant

- Investigated structural variation, genetic diversity, and evolution in humans and non-human primates using comparative and population genomic approaches.
- Led large, data-driven analyses of long-read sequencing data from major consortia (HGSVC, HPRC, T2T) and genetic biobanks spanning more than 5,500 samples.

- Awarded a Leakey Foundation Fellowship to generate and analyze sequencing data from four new gorilla haplotypes, characterizing their implications for health and evolutionary origins.
- Uncovered previously unresolved structural variation signals across temporal and geographic scales by applying population genetic packages and original computational methods.
- Managed high-performance cluster computing environments for large-scale analysis, with particular attention to handling confidential and protected datasets.
- Built reproducible, scalable Snakemake pipelines to automate data processing and analysis, readily translatable to new genomic regions and species.

Genentech Biological Technologies Intern, Genentech, South San Francisco

Jun 2019 – Sep 2019

Supervisor: Brian Murphy

- Built computational pipelines in R, Tableau, and Power BI to digitize two decades of assay data, cutting manual data-entry time and error rates.
- Developed an R Shiny app to automate paperwork and template filing, saving time and reducing administrative overhead for the research team.
- Selected for gLinux, a highly selective national 20-person cohort of undergraduate and graduate trainees, for exposure to industry and postgraduate research.

Canary Center at Stanford for Early Cancer Detection Intern, Stanford University, Palo Alto Jun 2018 – Sep 2018

Advisor: Dr. Ahmed El Kaffas

- Designed, fabricated, and prototyped a “flow phantom” device to study dynamic ultrasound microbubble behavior, with applications to medical imaging and targeted drug delivery.
- Developed a cost-efficient method using spun sugar (“cotton candy”) to mimic tumor vasculature, reducing material and personnel costs.
- Supported radiological data collection and analyzed ultrasonication devices in R, MATLAB, and Python to inform improvements in patient outcomes.
- Researched the emerging field of “radiogenomics”; literature review awarded 1st place, Norma J. Lang Prize for Information Research.

Peer-Reviewed Publications

Sridharan, S., Lou, R.N., Ferguson, S., et al. (2026). Recurrent structural variation and recent turnover at the 17q21.31 locus in humans and great apes. *Nat Commun*, 1(2), 123–123. <https://doi.org/10.1038/s41467-026-73174-1>

Preprints and Manuscripts in Preparation

Sridharan, S., Ferguson, S., Sudmant, P.H. [Manuscript in preparation].

El Kaffas, A., Gong, R., Sridharan, S., et al. (2020). Clinical Evaluation of Real-Time Optical-Tracking Navigation and Live Time-Intensity Curves to Provide Feedback During Blinded 4D Contrast-Enhanced Ultrasound Imaging. <https://doi.org/10.48550/arXiv.2011.00744>

Fellowships, Awards, and Honors

Parker Travel Award Finalist, UW-Genome Sciences Retreat	2026
NHGRI Genome Training Grant (T32), UW-Genome Sciences	2026
Waterson Travel Award, UW-Genome Sciences	2026
Trainee Travel Award, Long Read Sequencing Workshop	2026
Leakey Foundation Research Grant for Advanced Doctoral Students	2023
National Science Foundation Graduate Research Fellowship Program (NSF-GRFP) Honorable Mention	2022
UC Davis Outstanding Performance Citation, Departmental Citation, and Distinguished Scholar Award	2020
1st Place Norma J. Lang Prize for Undergraduate Information Research	2019
Aggie Alumni Advancing STEM Award	2019
Genentech-USBTD Outstanding Student in Molecular and Cell Biology	2018

Selected Conference Presentations

<i>Oral Presentation</i> , Long Read Sequencing Workshop, Jackson Laboratories, Farmington, CT	May 2026
<i>Poster Presentation</i> , Long Read Sequencing Workshop, Jackson Laboratories, Farmington, CT	May 2026
<i>Oral Presentation</i> , Bay Area Population Genetics, Stanford University	Dec 2025
<i>Oral Presentation</i> , Center for Computational Biology Retreat, Los Altos, CA	Oct 2025
<i>Oral Presentation</i> , Genetics, Genomics, Evolution, and Development Retreat, Berkeley, CA	Oct 2025
<i>Poster Presentation</i> , Biology of Genomes, Cold Spring Harbor Laboratory, New York	May 2025
<i>Poster Presentation</i> , Bay Area Population Genetics, University of California – Berkeley	Nov 2024
<i>Oral Presentation</i> , Society for Molecular Biology and Evolution (SMBE), Puerto Vallarta, Mexico	Jul 2024
<i>Poster Presentation</i> , American Society of Human Genetics (ASHG), Washington D.C	Nov 2023
<i>Poster Presentation</i> , Bay Area Population Genetics, University of California – Davis	May 2023
<i>Poster Presentation</i> , Biology of Genomes, Cold Spring Harbor Laboratory, New York	May 2023
<i>Poster Presentation</i> , Computing Research Association Widening Participation (CRA-WP), New Orleans, LA	Apr 2022

Teaching Experience

Graduate Student Instructor, MCELLBI 104 — Genetics, Genomics and Cell Biology, University of California, Berkeley Fall 2022, Spring 2025, Fall 2025

- Facilitated two weekly discussion sections (~25 students each) using chalk talks, slide presentations, and interactive polling.
- Developed discussion materials, including worksheets and chalkboard activities, tailored to individual, pair, and small-group work.
- Wrote and reviewed exam questions to align with section topics and learning objectives.
- Held twice-weekly office hours providing academic mentorship and guidance on research and career paths.
- Collaborated with fellow GSIs and instructors to refine lecture content and develop effective practice problems.

Student Advising

Alicia (Ally) Aggarwal, Summer Undergraduate Research Volunteer, Cornell University Jun 2026 – Present

Emily Liao, Undergraduate Research Shadowing Program, UC Berkeley Oct 2024 – Dec 2024

Community Engagement

Blog Editor-in-Chief, Berkeley Science Review Nov 2021 – Dec 2023

Writer and Editor, Berkeley Science Review Jul 2020 – Sep 2024

- Set content strategy for the Berkeley Science Review blog and print magazine, aligning topics with the publication's mission and its 2,000+ online and print readers.
- Directed a team of 10+ editors, maintaining rigorous standards for research quality and reader engagement while translating complex science for a broad, non-specialist audience.
- Authored 10+ articles ranging from 50-word short pieces to 2,000-word features on machine learning, evolution, and human health. Portfolio: berkeleysciencereview.com/author/samvardhini-sridharan

Python Boot Camp Co-Director, Center for Computational Biology, UC Berkeley Jan 2024 – Jun 2025

Python Boot Camp Lecturer and Teaching Assistant Jan 2022 – Jun 2022

- Developed and refined teaching materials on Pandas and NumPy, designing scaffolded examples to meet learners' evolving needs.
- Delivered lectures to a cohort of 200 students across in-person, virtual, and hybrid formats, tailoring hands-on exercises to a range of bioinformatics skill levels.

Service

Center for Theoretical and Evolutionary Genomics (CTEG) Organizing Committee	2023 – 2025
Center for Computational Biology Seminar Series Organizing Committee	2022 – 2023

Last updated: September 27, 2026